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p20250027

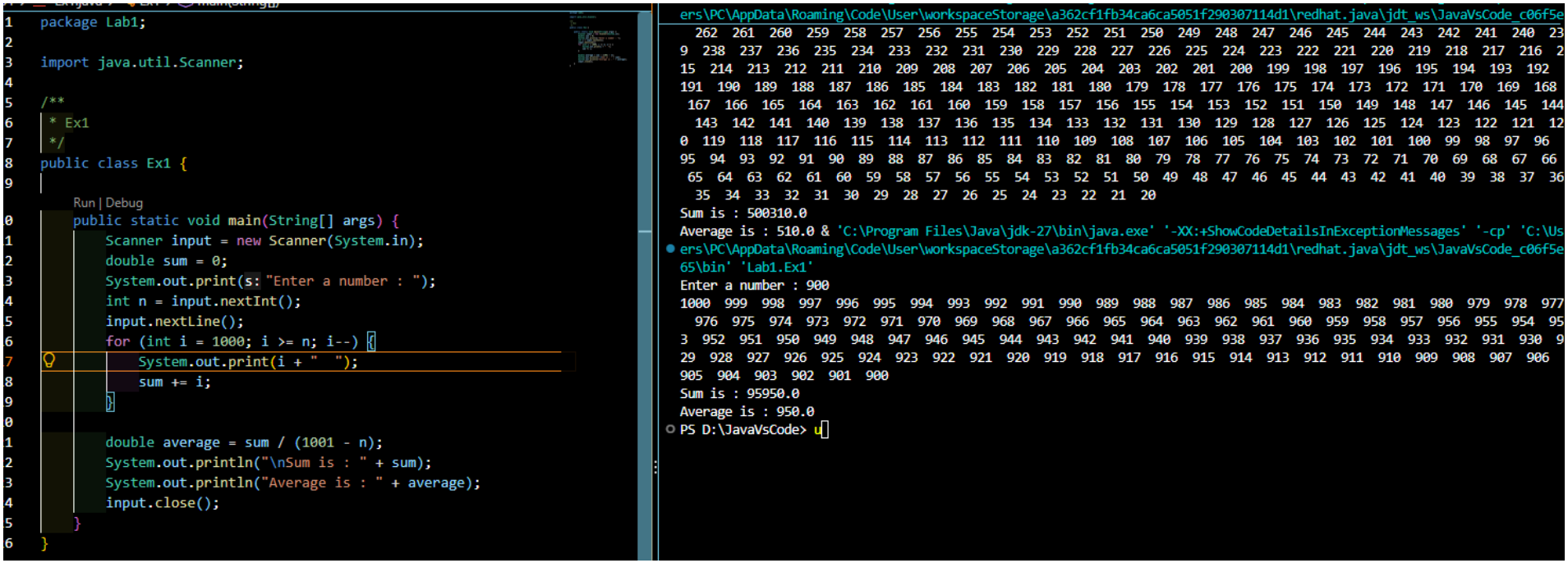
group A

Lab01: Basic Java syntax

I2-IP-AIECS

Lab1-Ex1:

SOLUTION :



```java
package Lab1;

import java.util.Scanner;

/**
 * Ex1
 */
public class Ex1 {

    Run | Debug
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        double sum = 0;
        System.out.print(s: "Enter a number : ");
        int n = input.nextInt();
        input.nextLine();
        for (int i = 1000; i >= n; i--) {
            System.out.print(i + "  ");
            sum += i;
        }

        double average = sum / (1001 - n);
        System.out.println("\nSum is : " + sum);
        System.out.println("Average is : " + average);
        input.close();
    }
}
```

```
ers\PC\AppData\Roaming\Code\User\workspaceStorage\a362cf1fb34ca6ca5051f290307114d1\redhat.java\jdt_ws\JavaVsCode_c06f5e
  262  261  260  259  258  257  256  255  254  253  252  251  250  249  248  247  246  245  244  243  242  241  240  23
9  238  237  236  235  234  233  232  231  230  229  228  227  226  225  224  223  222  221  220  219  218  217  216  2
15  214  213  212  211  210  209  208  207  206  205  204  203  202  201  200  199  198  197  196  195  194  193  192
191  190  189  188  187  186  185  184  183  182  181  180  179  178  177  176  175  174  173  172  171  170  169  168
 167  166  165  164  163  162  161  160  159  158  157  156  155  154  153  152  151  150  149  148  147  146  145  144
  143  142  141  140  139  138  137  136  135  134  133  132  131  130  129  128  127  126  125  124  123  122  121  12
0  119  118  117  116  115  114  113  112  111  110  109  108  107  106  105  104  103  102  101  100  99  98  97  96
95  94  93  92  91  90  89  88  87  86  85  84  83  82  81  80  79  78  77  76  75  74  73  72  71  70  69  68  67  66
 65  64  63  62  61  60  59  58  57  56  55  54  53  52  51  50  49  48  47  46  45  44  43  42  41  40  39  38  37  36
  35  34  33  32  31  30  29  28  27  26  25  24  23  22  21  20
Sum is : 500310.0
Average is : 510.0 & 'C:\Program Files\Java\jdk-27\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Us
ers\PC\AppData\Roaming\Code\User\workspaceStorage\a362cf1fb34ca6ca5051f290307114d1\redhat.java\jdt_ws\JavaVsCode_c06f5e
65\bin' 'Lab1.Ex1'
Enter a number : 900
1000  999  998  997  996  995  994  993  992  991  990  989  988  987  986  985  984  983  982  981  980  979  978  977
  976  975  974  973  972  971  970  969  968  967  966  965  964  963  962  961  960  959  958  957  956  955  954  95
3  952  951  950  949  948  947  946  945  944  943  942  941  940  939  938  937  936  935  934  933  932  931  930  9
29  928  927  926  925  924  923  922  921  920  919  918  917  916  915  914  913  912  911  910  909  908  907  906
905  904  903  902  901  900
Sum is : 95950.0
Average is : 950.0
PS D:\JavaVsCode> u
```

Lab1-Ex2:

SOLUTION:

Lab1 > Ex2.java > Ex2 > main(String[])

```java
package Lab1;

import java.util.Scanner;

public class Ex2 {
    Run | Debug
    public static void main(String[] args) {
        System.out.println(x: "================================");
        System.out.println(x: "Welcome to our program\n 'Quadratic Equation ax\u00B2' + bx +c =0 ");
        System.out.println(x: "================================");
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Input Value a:  ");
        double a = sc.nextInt();
        System.out.print(s: "Input Value b :");
        double b = sc.nextInt();
        System.out.print(s: "Input Value C:  ");
        int c = sc.nextInt();
        double delta = b * b - 4 * (a * c);
        System.out.println(x: "--------------------------------");
        System.out.println("Roots of this equation :" + a + "x\u00B2 +" + b + "x +" + c + " =0 are :\n");
        if (a == 0) {
            double x_1 = (-c) / b;
            System.out.println("X_1:" + x_1);
        } else {

            if (delta == 0) {
                double x_1 = (-b) / (2 * a);
                System.out.println("X_1 = X_2 : " + x_1);
            } else if (delta > 0) {
                double x_1 = (-b + Math.sqrt(delta)) / (2 * a);
                double x_2 = (-b - Math.sqrt(delta)) / (2 * a);
                System.out.println("X_1 : " + x_1);
                System.out.println("X_2 : " + x_2);
            } else {
                double x_1 = Math.sqrt(-delta);
                String formatted = String.format(format: "%.2f", x_1);
                System.out.println("X_1 : " + "-" + b + "+" + formatted + " i " + "/" + 2 * a);
                System.out.println("X_2 : " + "-" + b + "-" + formatted + " i " + "/" + 2 * a);
            }
        }
        System.out.println(x: "----------------------------------------");
        sc.close();
    }
}

```

```
PS D:\JavaVsCode>  & 'C:\Program Files\Java\jdk-27\bin\java
DetailsInExceptionMessages' '-cp' 'C:\Users\PC\AppData\Roam
aceStorage\a362cf1fb34ca6ca5051f290307114d1\redhat.java\jdt
e65\bin' 'Lab1.Ex2'
================================
Welcome to our program
 'Quadratic Equation ax²' + bx +c =0
================================
Input Value a:  1
Input Value b :2
Input Value C:  4
--------------------------------
Roots of this equation :1.0x² +2.0x +4 =0 are :

X_1 : -2.0+3.46 i /2.0
X_2 : -2.0-3.46 i /2.0
----------------------------------------
PS D:\JavaVsCode>  d:; cd 'd:\JavaVsCode'; & 'C:\Program Fi
java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' '
Roaming\Code\User\workspaceStorage\a362cf1fb34ca6ca5051f290
\jdt_ws\JavaVsCode_c06f5e65\bin' 'Lab1.Ex2'
================================
Welcome to our program
 'Quadratic Equation ax²' + bx +c =0
================================
Input Value a:  1
Input Value b :4
Input Value C:  2
--------------------------------
Roots of this equation :1.0x² +4.0x +2 =0 are :

X_1 : -0.5857864376269049
X_2 : -3.414213562373095
----------------------------------------
PS D:\JavaVsCode>
```

Lab1-Ex3:

Solution:

```java

import java.util.Scanner;

public class Ex3 {
    Run | Debug
    public static void main(String[] args) {
        int[] num = new int[5];

        System.out.println(x: "Input :");
        Scanner input = new Scanner(System.in);
        for (int i = 0; i < 5; i++) {
            num[i] = input.nextInt();
        }
        input.close();
        int min = num[0];
        int max = num[0];
        System.out.print(s: "Amoung all the five input numbers ");
        for (int i = 0; i < num.length; i++) {
            System.out.print(num[i] + ",");
            min = Math.min(min, num[i]);
            max = Math.max(max, num[i]);
        }
        System.out.println(": \nMin is " + min);
        System.out.println("Max is " + max);

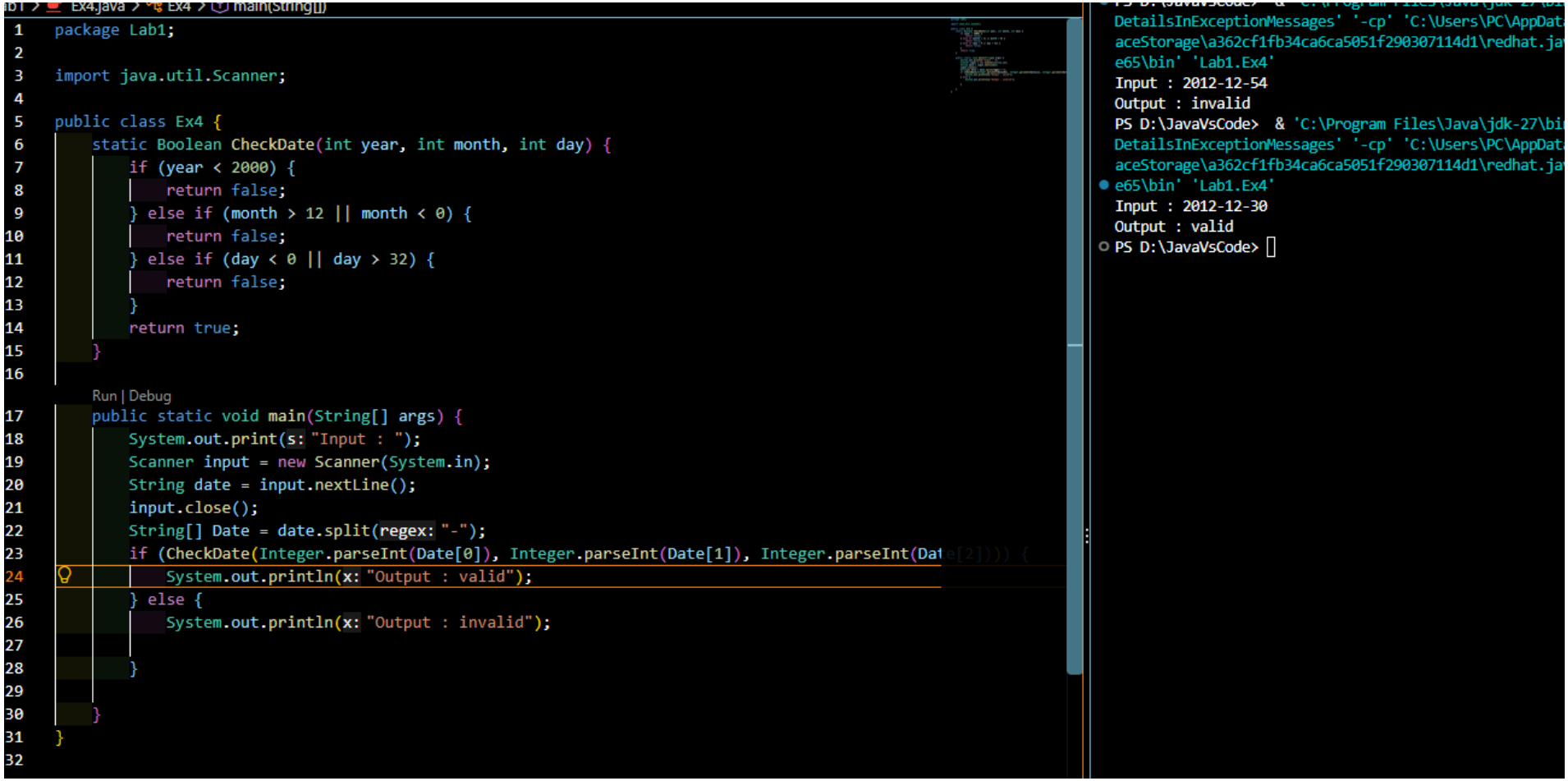
    }
}

```

```
aceStorage\a362cf1fb34ca6ca5051f290307114d1\redhat.java\jdt_ws\J
e65\bin' 'Lab1.Ex3'
Input :
12
10
20
13
20
Amoung all the five input numbers 12,10,20,13,20,:
Min is 10
Max is 20
PS D:\JavaVsCode>
```

Lab1-Ex4:

Solution:



```java
package Lab1;

import java.util.Scanner;

public class Ex4 {
    static Boolean CheckDate(int year, int month, int day) {
        if (year < 2000) {
            return false;
        } else if (month > 12 || month < 0) {
            return false;
        } else if (day < 0 || day > 32) {
            return false;
        }
        return true;
    }

    Run | Debug
    public static void main(String[] args) {
        System.out.print(s: "Input : ");
        Scanner input = new Scanner(System.in);
        String date = input.nextLine();
        input.close();
        String[] Date = date.split(regex: "-");
        if (CheckDate(Integer.parseInt(Date[0]), Integer.parseInt(Date[1]), Integer.parseInt(Dat
            System.out.println(x: "Output : valid");
        } else {
            System.out.println(x: "Output : invalid");

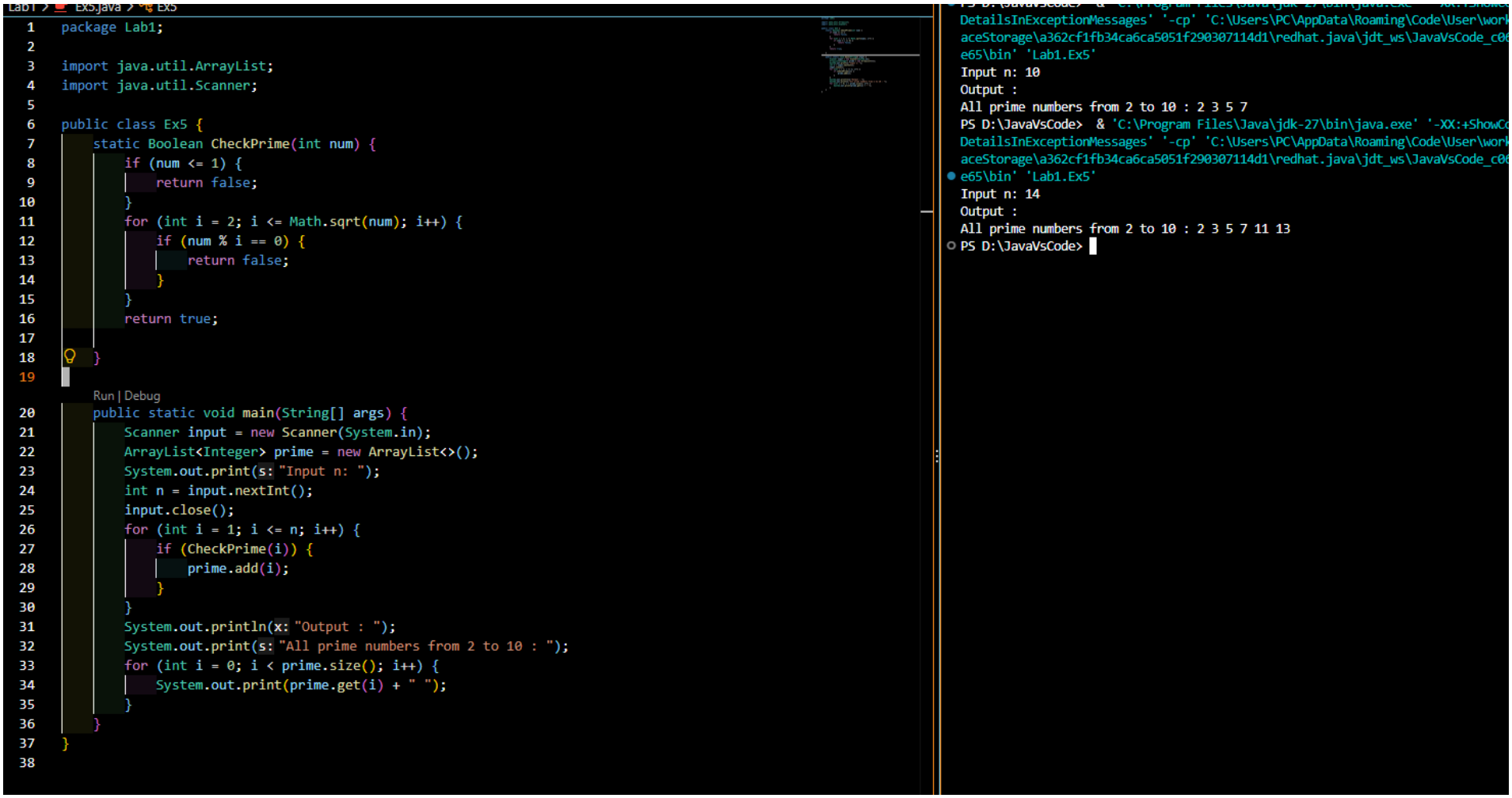
        }

    }
}
```

```
DetailsInExceptionMessages' '-cp' 'C:\Users\PC\AppDat
aceStorage\a362cf1fb34ca6ca5051f290307114d1\redhat.ja
e65\bin' 'Lab1.Ex4'
Input : 2012-12-54
Output : invalid
PS D:\JavaVsCode>  & 'C:\Program Files\Java\jdk-27\bi
DetailsInExceptionMessages' '-cp' 'C:\Users\PC\AppDat
aceStorage\a362cf1fb34ca6ca5051f290307114d1\redhat.ja
e65\bin' 'Lab1.Ex4'
Input : 2012-12-30
Output : valid
PS D:\JavaVsCode>
```

Lab1-Ex5:

Solution:



Lab1-Ex6:

Solution:

